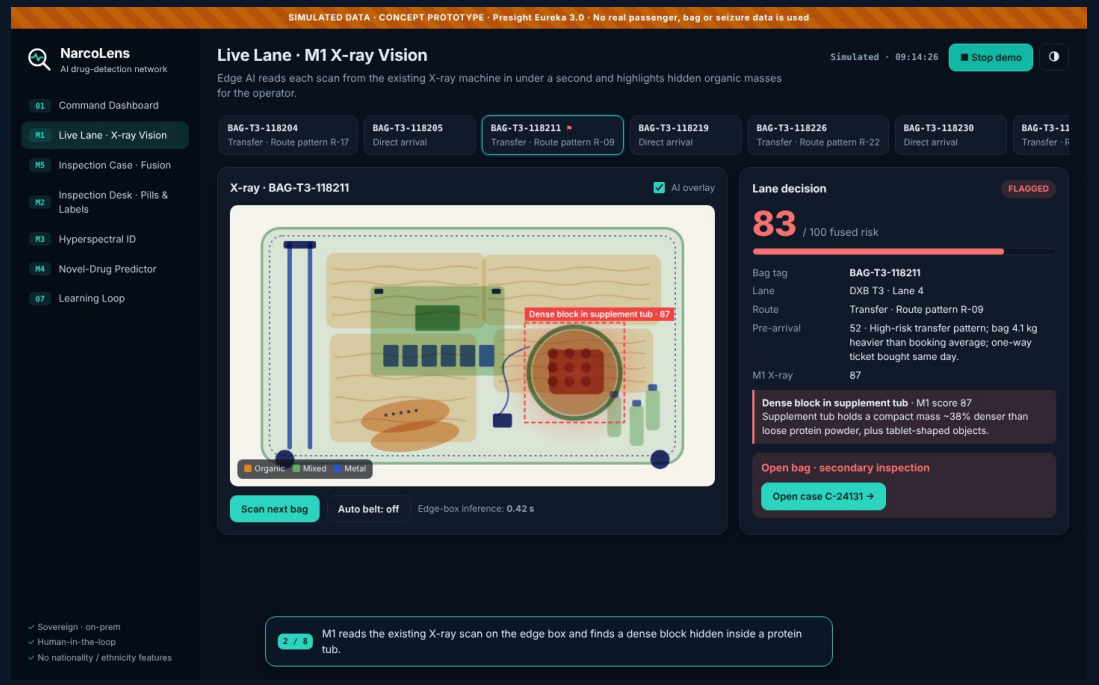
PRESIGHT EUREKA 3.0 · DEFENSE & POLICING

NarcoLens

Turning every UAE customs scanner into an AI drug detector

Software-only **Sovereign · on-prem** **Working prototype**

TRL 3 · Clickable prototype on simulated data is attached (web build + guided demo)



Prototype: M1 X-ray Vision flags a dense block hidden inside a protein tub

Smugglers are out-innovating manual screening

1,273

drug seizures at Dubai's land, sea and air ports in one year

540

drug cases at Dubai airports in nine months (Jan–Sep 2023)

14M

Captagon pills hidden in a single commercial shipment (Feb 2026)

Figures from Dubai Customs and press reports, 2023–2026

Hidden in plain sight

- Pills pressed into clothing buttons
- Drugs packed inside furniture and door panels
- Opium mixed into protein powder and shoe linings

Still found mostly by hand

- Operators judge X-ray density by eye, bag after bag
- Sniffer-dog teams cover limited hours and lanes
- Naming the substance waits days for the lab

The gap: detection hasn't kept pace

No AI reads drug X-rays

Operators judge density by eye. We found no published system that auto-flags narcotics at UAE checkpoints.

Chemistry needs a lab

Confirming a substance takes days, so officers act on suspicion alone at the desk.

Synthetic drugs mutate

New variants are designed to dodge detection libraries; detection lags behind the first seizure.

Parcels are booming

E-commerce and "research peptide" shipments grow faster than inspection capacity.

Why now: edge GPUs and hyperspectral cameras are affordable, the UAE is investing in sovereign AI, and customs data is already digital. The missing piece is software that connects it all.

NarcoLens: one AI layer, one case per bag

M5 · Pre-arrival risk

Route and booking patterns score each bag early

M2 · Pills, vials, labels

Imprint match, Arabic/English OCR, registry check

M1 · X-ray Vision

Edge AI reads the existing scan in under 1 s

INSPECTION CASE

one per bag tag
M5 fuses all evidence

M3 · Hyperspectral ID

Names the substance through the packaging

Loop · Federated learning

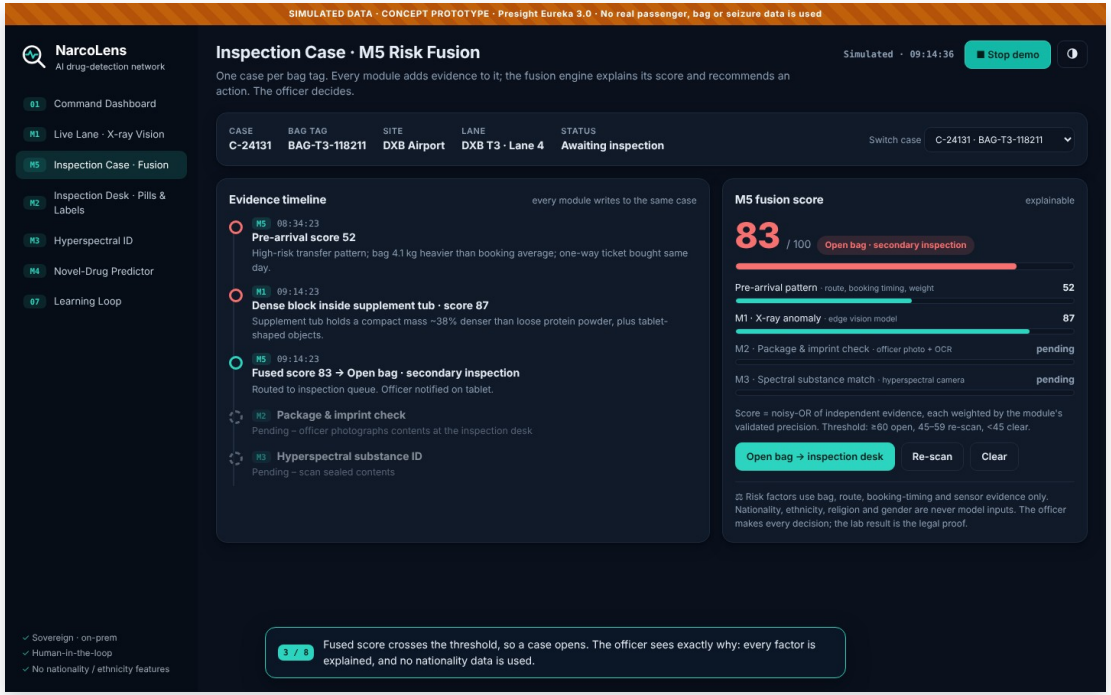
Each confirmed seizure retrains every site

M4 · Novel-drug predictor

Predicts new variants and feeds the M3 library

A bag's journey through NarcoLens

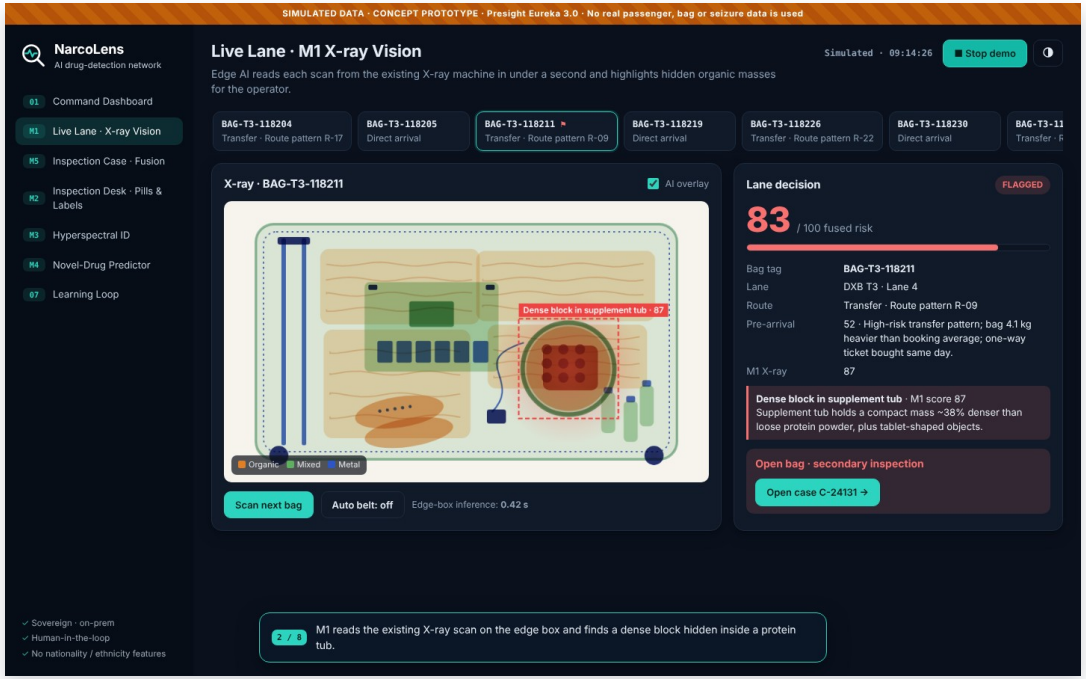
1. **Before landing · M5**
   Route and booking pattern score the bag 52/100
2. **On the belt · M1**
   Dense block in a protein tub flagged in 0.4 s (87)
3. **Case opened · M5**
   Fused score 83: open the bag, reasons shown
4. **Inspection desk · M2**
   Label not in registry; Captagon-style imprint
5. **Hyperspectral · M3**
   Amphetamine-type stimulant, 98% match, still sealed
6. **Lab confirms · Loop**
   Seizure confirmed; all 7 sites retrain, no images shared

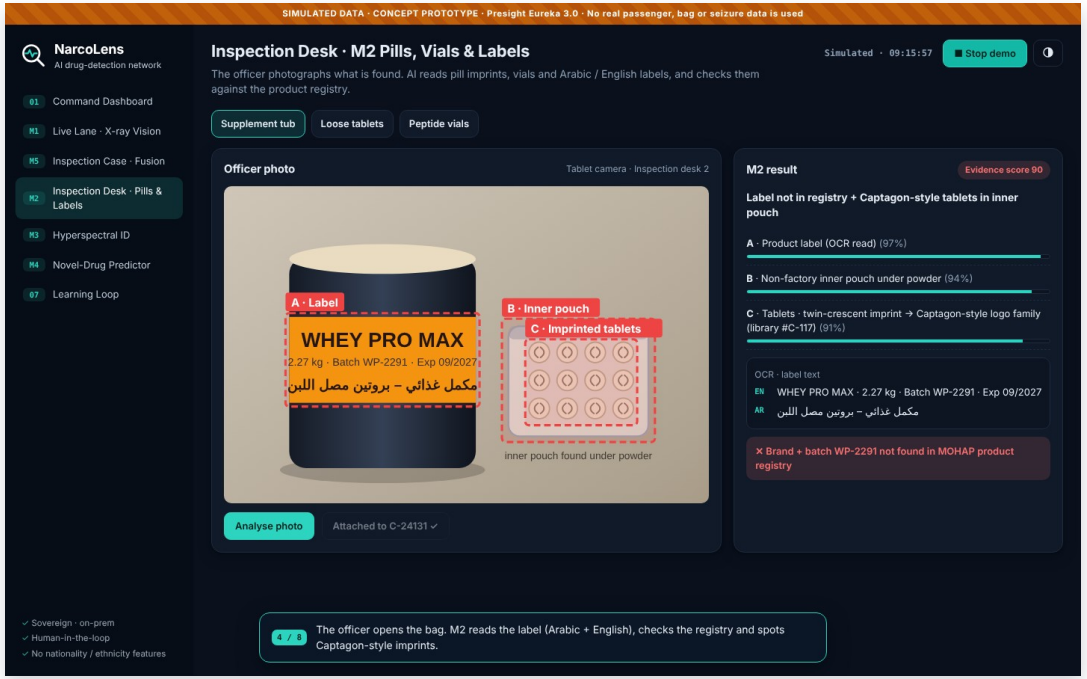


Prototype: every module writes to the same case, and the score explains itself

Under a minute from belt to substance ID in the demo, and no bag is opened until the evidence justifies it

Seeing what's hidden: the vision modules





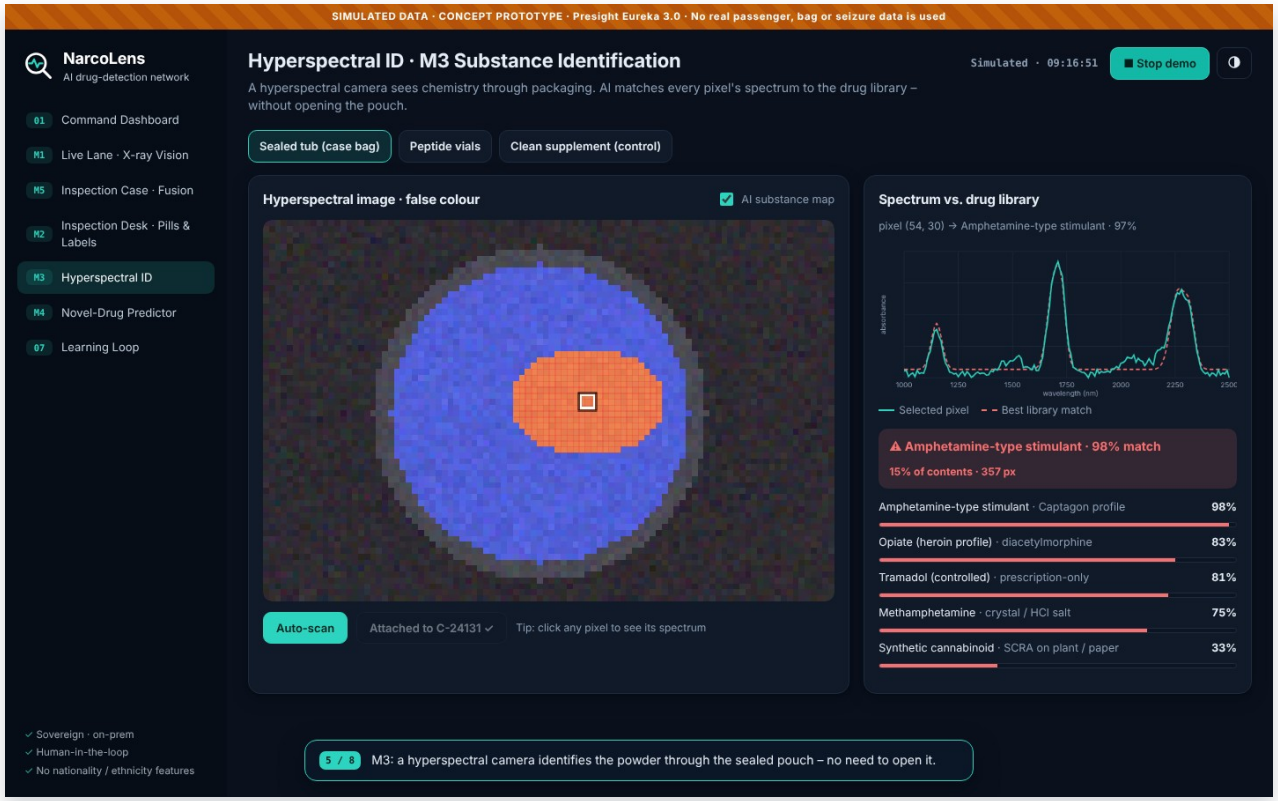
M1 · X-ray Vision

- Edge GPU beside the existing scanner, under 1 s per bag
- Finds organic masses, hidden cavities, abnormal density
- RT-DETR detector + PatchCore anomaly model

M2 · Pills, vials and labels

- Officer photographs what is found on a desk tablet
- Matches pill imprints and vials; reads Arabic/English labels
- Fine-tuned ViT/CLIP + Arabic OCR, checked against registry

Seeing chemistry: identify, then predict



Prototype: M3 finds amphetamine-type powder inside a sealed pouch (98% spectral match)

M3 · Hyperspectral ID

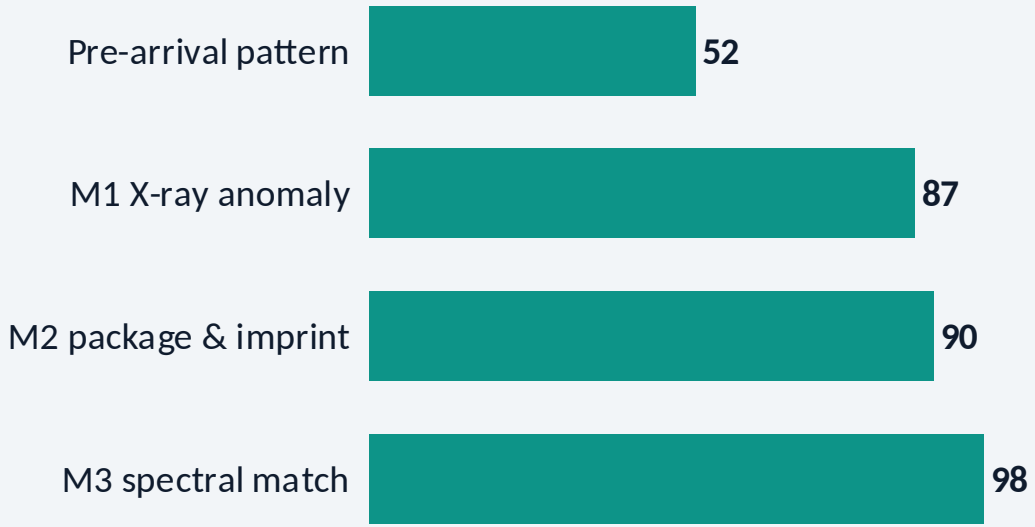
- Camera reads ~150 bands from 1,000 to 2,500 nm
- Every pixel matched to the drug library
- Names the drug without opening the package

M4 · Novel-drug predictor

- Graph neural network predicts new variants' spectra
- Chemists validate, then it reaches every camera
- Recognises a variant before its first seizure

Explainable by design, accountable by default

M5 fusion: why this bag scored 99



Each bar is independent evidence, weighted by the module's validated precision. Thresholds: 60+ open · 45–59 re-scan · under 45 clear.

No nationality or ethnicity

Only bag, route, timing and sensor evidence. Bias audits every quarter.

The officer decides

AI recommends; every search and seizure is a logged human decision.

The lab is the legal proof

AI evidence pack (images, spectra, scores) kept under chain of custody.

On-prem and federated

Models train at each site; zero raw images leave a site.

Software-only, vendor-agnostic architecture

1 · Edge

- GPU box beside each X-ray/CT scanner and hyperspectral camera
- Runs M1 and M3 locally in under 1 s
- Reads images via the DICOS standard

2 · Core (on-prem)

- Inspection cases and M5 fusion
- M4 predictor and drug library
- Federated aggregation, model registry, audit log

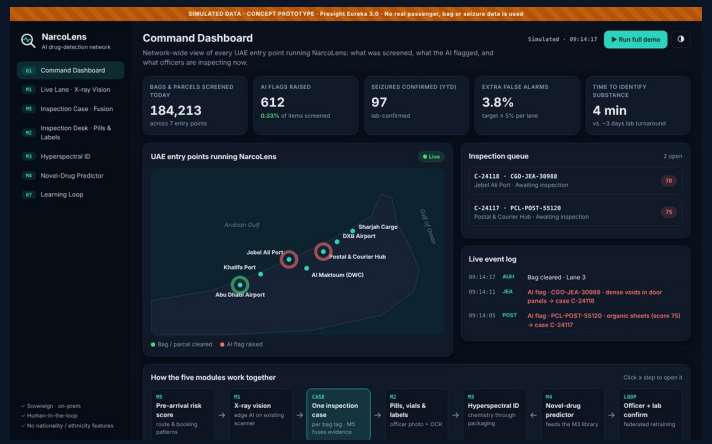
3 · Apps

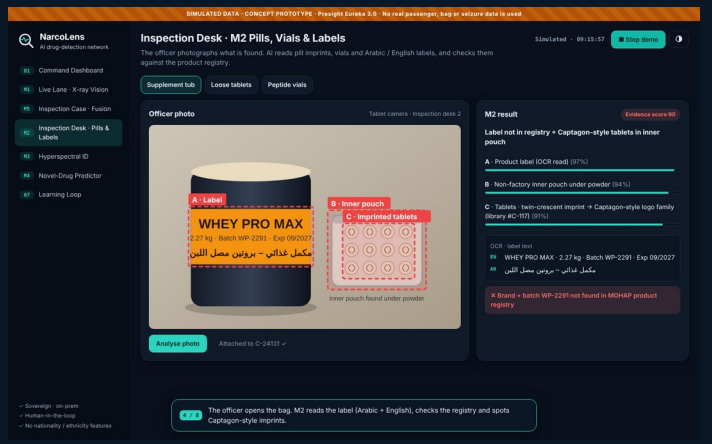
- Overlay on the scanner operator's screen
- Officer tablet app at the desk
- Command dashboard for supervisors

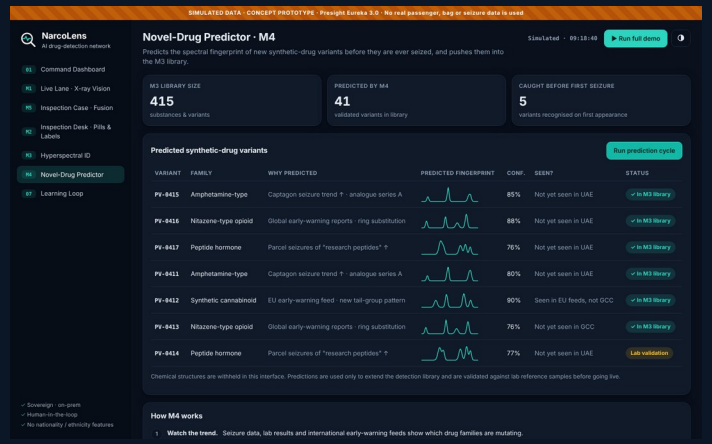
Integrates with
Any scanner vendor (DICOS) · customs manifest and booking systems · forensic lab systems

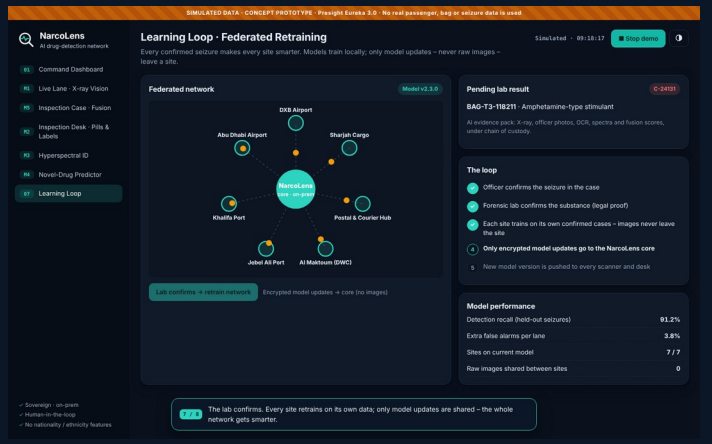
Built with
PyTorch · RT-DETR · PatchCore · ViT/CLIP · Arabic OCR · PyTorch Geometric · XGBoost · Flower · Kubernetes · React

Working prototype: try it now









Command dashboard · Inspection desk (M2) · Novel-drug predictor (M4) · Learning loop

What judges can do

1. Press "Run full demo": one bag end to end in 45 s
2. Scan bags and toggle the AI overlay (M1)
3. Analyse an officer's photo (M2)
4. Auto-scan, then click any pixel (M3)
5. Run a prediction cycle (M4)
6. Retrain the network after a seizure

All data simulated · runs offline in any browser · no login

Expected impact: pilot targets

+30%

seizures per 1,000 bags inspected

Minutes

to identify a substance, vs ~3 days in the lab

−40%

operator review time per flagged bag

≤5%

extra false alarms per lane

Targets to be validated in an 8-week shadow-mode pilot before any live use

Who benefits

- Dubai and Abu Dhabi Customs, ICP
- Police anti-narcotics units
- Honest travellers: fewer manual searches

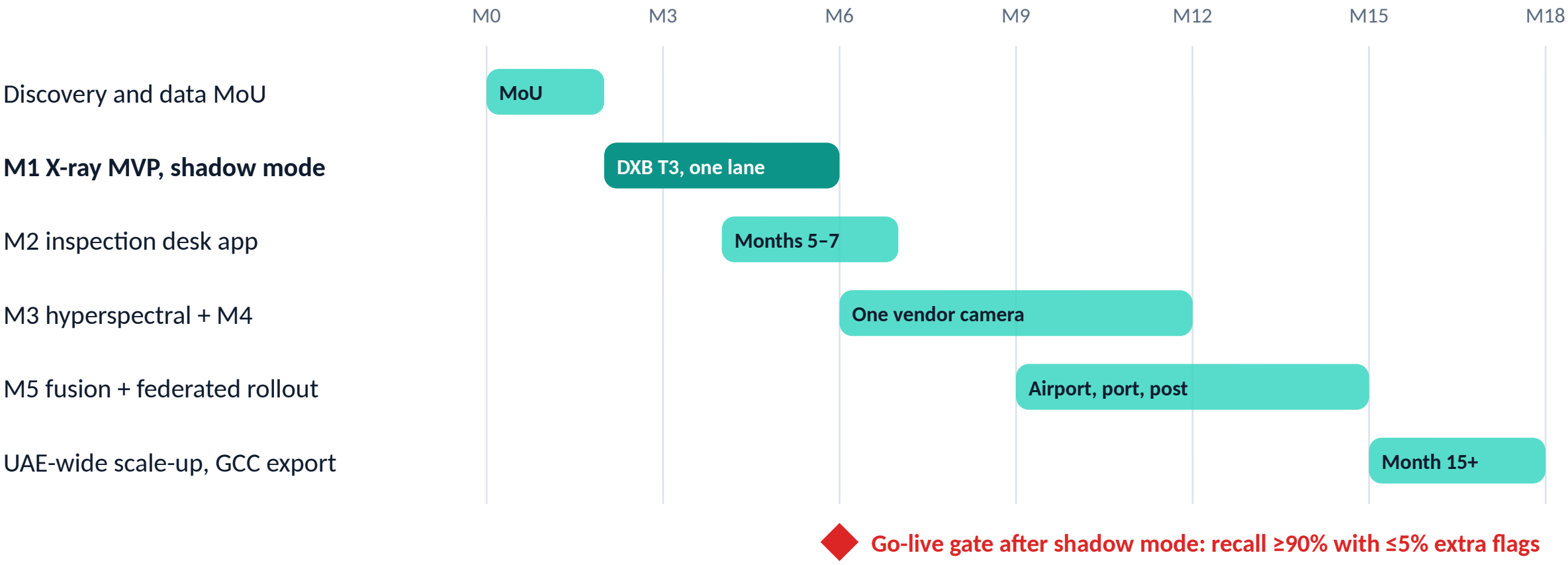
Value for Presight

- Sovereign product for UAE border security
- Export to GCC customs agencies
- Recurring licence and support per site

Beyond drugs

- Weapons and explosives
- Counterfeit goods and wildlife trafficking
- Undeclared currency and gold

Roadmap: shadow-mode pilot to UAE-wide rollout



Team at peak ≈ 12: 3 computer-vision engineers · 1 spectroscopy ML scientist · 2 data engineers · 2 MLOps · 2 full-stack · product lead with customs expert · responsible-AI and legal lead

Risks we've designed for

Risk	How NarcoLens handles it
Little labelled drug X-ray data	Threat-image projection, public X-ray pretraining, active learning from officer feedback
False alarms slow the queues	Shadow mode first; per-lane thresholds; officers see only top-risk bags
Closed scanner formats	DICOS standard; early partnership with one or two scanner vendors
Profiling and bias	No nationality or ethnicity features; quarterly bias audits; full audit log
Evidence in court	AI only flags; the lab test stays the legal proof; explainable heatmaps and chain of custody

Why Presight

- Sovereign, on-prem AI for UAE government is our core business
- Big-data analytics for public safety and investigations
- In-house talent across computer vision, Arabic NLP and MLOps

The team and the ask

Team

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

[Name · Role · Department]

Up to 5 Presight employees

Our ask

1. **An ELT sponsor**
   to champion NarcoLens with UAE border agencies
2. **An introduction to Dubai Customs and ICP**
   to agree a data-sharing MoU for historical scans
3. **A 3-month shadow-mode pilot**
   on one DXB Terminal 3 lane, with no impact on passengers

One seizure at DXB should make every UAE border smarter.